

MADHAV KOTAK

Vadodara, Gujarat | +91 8758249503 | madhavgotak124@gmail.com | [LinkedIn](#) | [GitHub](#)

Professional Summary

AWS Certified Final Year B.Tech CSE student with hands-on experience in cloud networking, infrastructure automation, CI/CD pipelines, and DevSecOps. Skilled in Python, Terraform, Docker, and GitHub Actions. Built multi-cloud governance and compliance solutions on AWS. Aspiring Cloud and DevOps Engineer.

Education

B.Tech – Computer Science & Engineering

Parul University, Vadodara | Sept 2023 – Present

CGPA: 8.56/10

Relevant Coursework: Computer Networks, Network Security, Cloud Computing, DSA, OS, DBMS

Technical Skills

Networking: TCP/IP, OSI Model, Subnetting & CIDR, DNS, NAT, Security Groups, NACLs

Cloud: AWS (EC2, S3, RDS, Lambda, IAM, CloudWatch, VPC, CloudFormation)

DevOps: Docker, GitHub Actions, Terraform, Git, Kubernetes

Programming: Python (Flask), Bash, SQL

Databases: PostgreSQL, MongoDB

OS: Linux, Windows

Concepts: Data Structures & Algorithms, Microservices, Virtualization

Projects

Multi-Cloud Governance Platform

Jan 2026 – March 2026 | [Live Demo](#) | [GitHub](#)

Tech Stack: Python, FastAPI, Terraform, Docker, GitHub Actions, AWS, JWT, PostgreSQL

- Engineered a containerized deployment governance platform that automates configuration approvals and compliance validation, reducing manual validation effort by ~70-80% using a policy-as-code YAML engine.
- Designed declarative Terraform IaC blueprints for cloud resource provisioning, integrating automated dry-runs (terraform plan); built a state-machine deployment control plane with complete event auditing and rollback support.
- Orchestrated a multi-service container environment using Docker Compose for local integration testing; built a FastAPI control plane with JWT authentication and RBAC to manage deployment configurations across logical tenant boundaries.
- Integrated GitHub Actions CI/CD pipeline for automated testing, linting, and deployment, achieving zero-downtime releases.

CloudSentinel – CSPM & DevSecOps Platform

April 2026 – May 2026 | [Live Demo](#) | [GitHub](#)

Tech Stack: Python, Flask, AWS Boto3, Checkov, Trivy, Docker, Kubernetes, JWT, PostgreSQL, Chart.js

- Developed live AWS CSPM auditor using Boto3 to evaluate VPC Security Group ingress/egress rules, IAM privilege escalation vectors, and S3 public policies against CIS benchmarks – reducing network compliance reporting time by 85%.
- Engineered Shift-Left DevSecOps workflow integrating Checkov and Trivy to automate pre-deployment static analysis of Terraform network configurations and container vulnerability screening, catching 100% of critical issues.
- Built Terraform drift comparator evaluating expected network state against live-deployed cloud assets, reducing reconciliation time by 90% through real-time diff timelines.
- Migrated the Docker-based container stack to a local Kubernetes (Kind) cluster, configuring PersistentVolumeClaims (PVC), ConfigMaps and RBAC policies for secure, inside-the-cluster scanning.

Certifications

- AWS Certified Cloud Practitioner – Amazon Web Services | [View Badge](#)
- AWS Academy Cloud Foundations – Amazon Web Services | [View Badge](#)

Achievements & Awards

- Selected in Top 45 teams out of 720+ teams at Vadodara Hackathon 6.0 (2025)
- Solved 200+ questions on [LeetCode](#)